

On-Chip Sensor to Monitor Aging Evolution in FinFET-Based Memories

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Abstract — Aging of electronics can be caused by various physical mechanisms, including bias temperature instability (BTI), hot carrier injection (HCI), time-dependent dielectric breakdown (TDDB) and electromigration (EM). Aging phenomena may lead to a degradation of performance and reliability of an electronic system, hence limiting its expected lifetime. The goal of this paper is twofold. First, it presents an advanced, miniature design of an aging sensor; second, it shows how this sensor is integrated with a commercial 5nm FinFET-based SRAM IP core. The sensor is implemented with a small number of logic gates, resulting in negligible overhead in area, power and delay. The sensor is also equipped with power gating capability. Thus, during idle time, the sensor is temporarily disconnected from the power supply lines to prevent aging of its transistors. Experimental results based on extensive SPICE simulations demonstrate that this approach is able to detect even the slight increase in memory response time due to aging much earlier than a functional error is observed at the memory output. This solution enables mission-mode monitoring of memory operation, which is a critical aspect of silicon lifecycle management (SLM) framework.

Index Terms — aging monitoring, on-chip sensor, reliability, FinFET-based memory, silicon lifecycle management (SLM).

I. INTRODUCTION

Device scaling down increases sensitivity to parametric variations. This requires wider safety margins in the design corners that consider *worst case PVT* (Process, supply Voltage and Temperature) variations. Furthermore, for long life operation products, *aging effects* also limit performance [1].

Simulation tools predict conservative values for maximum clock frequencies (fclk), which leads to performance waste [2]. In fact, in real products, either indicated fclk can be achieved with lower power consumption, or greater values of fclk can be used with typical core VDD. Thus, if *on-chip aging evolution* can be monitored, significant gains can be achieved in performance or power, even for safety-critical products.

Several approaches can be found in the literature, where on-chip sensors have been deployed to *locally* monitor system performance. The approach described in [3] and [4] can be implemented to monitor combinational logic either in FPGA or ASIC. In [5], authors implemented this sensor in a commercial FPGA and performed aging measurements for several aging states of the component.

Even though authors do not propose any new on-chip sensor architecture in [6], they present an interesting approach based on workload analysis to identify the most critical timing-critical flip-flops in a combinational logic

(i.e., those FFs the most exposed to static BTI). Then, they claim they can minimize the overheads resulting from the inclusion of sensors in the functional circuit. This approach is conceived for ASIC, but under certain constraints, it can be applied to FPGA-based designs.

In [7], authors proposed a low-cost sensor for aging and late transitions detection for implementation in FPGAs. It is an interesting approach, but it is worth noting that the type of required D flip-flops (DFFs), control (clock) signals and detection window are of complex implementation. Moreover, since the data yielded at the critical path output is also used to control the clock signal of one of the sensor's DFFs, the approach is sensitive to glitches along large functional blocks. These parameters are not easy to control, in particular if we take into consideration PVT variations, which in practice, limits the applicability of the approach.

In [8], authors present an interesting work where they review a number of unique test chip designs pursued by circuit designers that demonstrate the benefits of utilizing on-chip logic and a simple test interface to automate circuit aging experiments.

A. On-chip sensors for SRAM

Aging phenomenon negatively impacts the functional characteristics of SRAMs, since it degrades speed, operating voltage, memory cell's noise margin and sense amplifier's input offset voltage. Ultimately, functional failure is observed. Traditionally, designers use guard bands in several forms; extra margins are considered to guarantee that the memory will operate correctly during its lifetime. However, this approach negatively impacts the performance of the circuit since it affects speed, power consumption, area and possibly the yield. Thus, it is imperative to develop aging-tolerant design techniques that will provide the ability to sense aging levels, predict upcoming failures in the memory and early react to retain the reliable operation.

In [5] and [9], authors presented an on-chip aging sensor monitor off-line writing operations into the SRAM memory cells in order to detect degradation. The approach was evaluated on a 65nm CMOS SRAM memory array. A dedicated sensor (which is based on a sense amplifier topology) exists for every column in the SRAM array. The technique assumes the use of power gating schemes and the existence of a non-aged reference column of memory cells. Off-line write operations are performed to the memory cells and the corresponding reference cells. After each write operation, the virtual VDD voltage levels of the memory cell under monitoring as well as the reference cell are compared in order to detect aged memory cells. The main

drawbacks of this approach are: (a) aging effects on periphery logic (e.g., read/write circuitry, line/column decoders, pre-charge, equalizer circuitry and sense amplifiers) are not detected, and (b) there is a need of non-aged reference cells.

In [10], authors presented the integration of a built-in current sensor (BICS) in the design of a 28nm SRAM IP core. The goal is to detect aging of the array of cells. Unfortunately, faults affecting periphery logic, similarly to [5] and [9], are not detected. The same comments can be tailored for the work published in [11] and [12].

IDDQ current has been used in [13] for the characterization of NBTI-induced SRAM performance degradation. Under NBTI influence, the leakage current of the memory cells is reduced exponentially due to the V_t degradation. The presence of a power-gating scheme is assumed, and a current mirror-based IDDQ monitor is exploited. This approach measures the accumulative leakage current of the whole memory array and thus the location of aged cells is not feasible. In other words, it is not possible to discriminate the aging state of each cell sitting in the memory array, so it is not possible to predict functional failure unless very conservative decisions are taken. Moreover, aging effects on periphery logic are not detected.

In [14], authors exploit special structures that are based on analog blocks (voltage comparators, phase comparators, etc.) to characterize the stability of an SRAM under the influence of NBTI (among other sources of vulnerability). However, the design complexity is too high.

In this context, the goal of this paper is twofold. First, it presents an advanced, tiny aging sensor design, and second, it integrates this sensor with a commercial 5nm FinFET-based SRAM. The sensor was implemented with a very small number of logic gates, which results in negligible area, power and delay overheads. The sensor is connected at the end of delay-critical paths of functional blocks in parallel with the original flip-flops sitting at these points. Experimental results based on extensive SPICE simulations demonstrate the approach is able to detect slight delay increases in the memory time response due to aging, much earlier than a functional error can be observed at the memory output. By functional error, we understand that the expected value could not be read at the memory output, no matter the reason was (e.g., increased time response due to aging of the memory cell, line and column decoders, write logic, or sense amplifiers). Moreover, any electromigration effect on long bit-lines that result in memory response time increase is liable to be detected by the proposed approach. This solution allows performing mission-mode monitoring of memory operation, which is a crucial aspect of silicon lifecycle management (SLM) framework.

B. Comparison with known solutions

We have compared our aging sensor circuit with other sensors presented in the literature in terms of area, technology node, application and configurability. As for area overhead, it has been roughly estimated by considering standard cell areas used in sensor design.

We acknowledge that comparison might be not accurate, since the other sensors are implemented in different technology nodes. However, as shown in Table I, the clear advantage of proposed sensor is the ability to configure the security margin (SM) window as well as low area consumption and full digital implementation.

TABLE I. Comparison between different sensors

Sensor Type	Area (μm^2)	Technology Node (nm)	Fixed SM window	NBTI Only	Appl.
[15]	485	65	YES	YES	Analog
[16]	60	45	YES	NO	Digital
Our	0.58	5	NO	NO	Digital

Another key advantage of the proposed aging sensor is that sensor itself is not affected by aging, as it is powered on only during measurement cycles, thus assuring high accuracy.

The paper is organized as follows: Section II describes the proposed aging sensor; Section III presents the architecture and functionality of 5nm SRAM IP core as well as the connection of the proposed sensor at the output of such memory block; Section IV describes the SPICE electrical simulations and discusses the obtained results. Finally, Sections V and VI summarize the conclusions and future work.

II. PROPOSED AGING SENSOR

We are investigating various on-chip aging sensors for monitoring system's health, so that to allow silicon lifecycle management (SLM) [17]. One of the sensors that has been implemented is shown in Figs. 1 and 2. This sensor is an adaptation of the one first proposed by [3], [4], where the sensor was used to monitor aging evolution of combinational logic (with case-study implemented in a commercial FPGA). More specifically, we simplified the sensor design presented in [3] and [4] based in two aspects: (a) by reducing the number of gates required to implement the sensor, and (b) by proposing a more robust, comprehensive set of clock signals to control the sensor once it is embedded in the functional logic. Two basic advantages arise when compared this sensor with others found in the literature: (a) its low hardware complexity and ease control renders it a robust and very attractive solution; and (b) can be either implemented in ASIC or FPGA. As observed in Fig. 1, the sensor is placed at the end of delay-critical paths of functional blocks (in parallel with the original DFF sitting at this point). In Fig. 1, the signals in orange: *Clk_sens*, *Clk_ei* and *EI* are signals added (with respect to the original DFF) to have full control of the sensor. *Clk_sens* is used to control the sensor by generating a synchronous clock with respect to the original clock (*Clk_ori*) of the functional circuit (but phased in advance with respect to the original clock, as described later). *Clk_ei* updates the sensor output with the error indication status (*EI*).

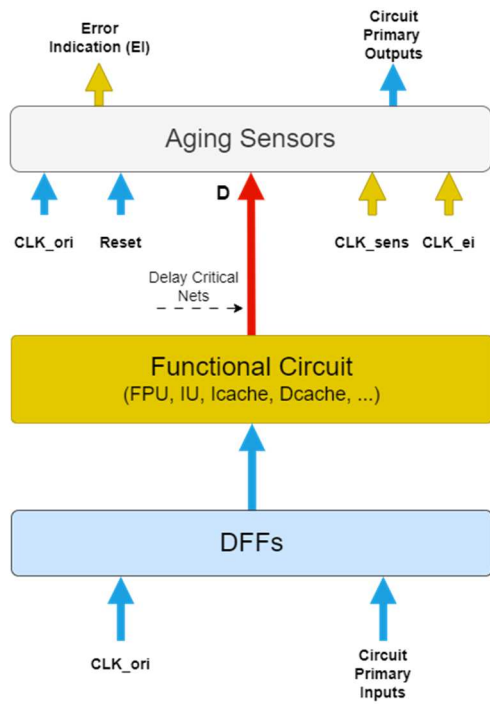


Fig. 1. Aging sensor placement.

Fig. 2 depicts the internal logic of the sensor. *Clk-ori* is the original clock of the functional block, *Clk_sens* is the clock of the sensor, *Data* is the data issued by the delay-critical path of the functional block, *reset* is the reset signal which is activated before the functional block starts operating and every time the user wishes to reset the error indication output (EI) of the sensor. Finally, *out_dff* is the data output of the critical path.

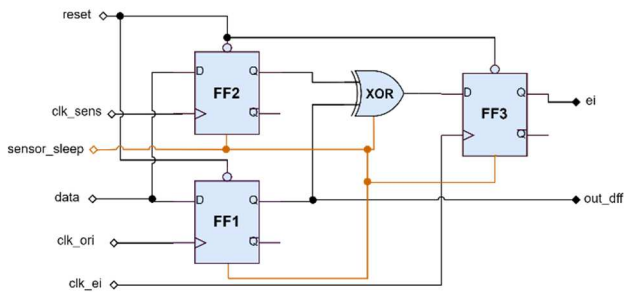


Fig. 2. Aging sensor: internal logic.

The sensor is featured with power gating capability (Fig. 3) that aims to reduce the effect of aging on sensor itself by selectively waking it up during measurement cycles. During idle times the sensor is temporarily disconnected from power supply lines to prevent its transistors from aging. Above feature is implemented by applying so called fine-grain power gating technique, when regular standard cell logic is modified with an additional NMOS switch transistor to gate VSS rail, thus eliminating leakage current flowing from VDD to VSS during sleep mode.

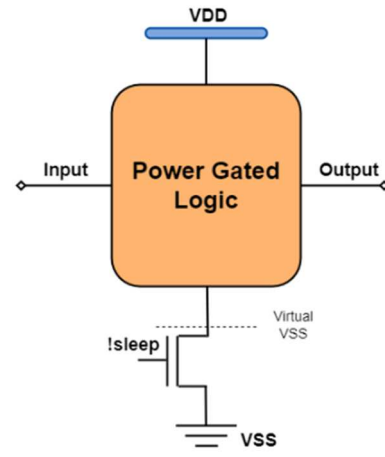


Fig. 3. Principle of applied Fine-Grain Power Gating

Fig. 4 illustrates the aging principle and the timing violation detection process. The approach works as follows: it is defined a security margin (SM) as the time interval in which the data signal propagated through the critical path becomes stable (the so called, steady state) where no more signal transitions are observed in this time period. The sensor is controlled by the *Clk-sens* signal, which is activated at the very beginning of the SM time window. Thus, any increase in signal propagation delay that drives the last transition of this signal into the SM is liable to be detected by the sensor.

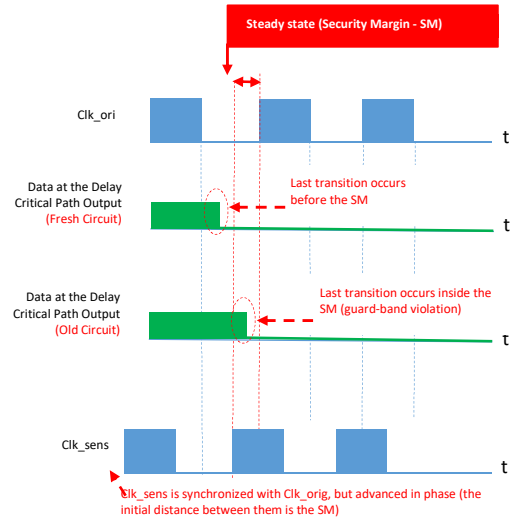


Fig. 4. Aging principle and timing violation detection.

Fig. 5 depicts the hierarchical integration of the sensor into a large circuit. Currently, we are integrating 64 sensors in a RISC-V processor. These sensors are connected to the most-significant bits of the floating-point multiplier (FPMul) block and to the outputs of the L1 cache memory. In both cases, the sensors are placed in parallel with the output DFFs connecting these functional blocks to the processor bus, and deal to monitor aging evolution on the critical paths of the FPMul and on the response time of the memory. The sensors also monitor the delay increase provoked by some cases of electromigration in the cache bit lines and in FPMul tracks. The sensors have been located in

these specific blocks because they are among the most timing sensitive nodes of the processor, i.e., the nodes operating with the narrowest clock time slack (so, the nodes with the highest probability to fail as aging evolves during chip in-field operation). The RISC-V processor was designed with the IHP mid-range 130 nm SiGe BiCMOS rad-hard technology.

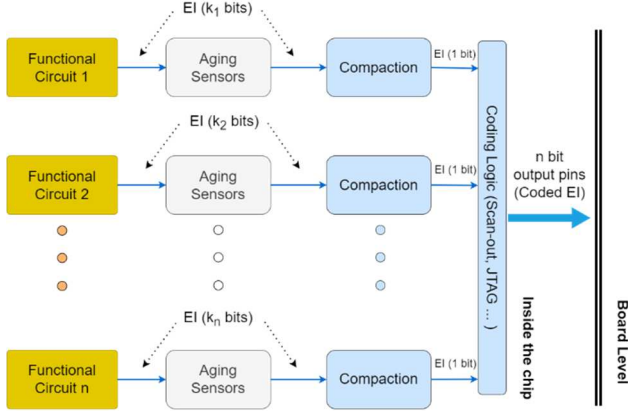


Fig. 5. Sensor integration at the system level.

III. APPLICATION OF AGING SENSOR ON FINFET-BASED SRAM

In order to test the capabilities of proposed aging sensor to detect aging of advanced node memory IPs, an experimental setup is introduced which includes a 5nm FinFET-based SRAM IP core with all peripheral circuitry, namely decoder, cell driver, sense amplifiers, pre-charge and equalizer circuitry. For this experiment an internal Advanced Inductive Failure Analysis (AIFA) methodology is used [18].

Aging sensors are positioned in parallel with the memory output registers that connect the memory to the system bus (just after the sense-amplifiers output nodes). This positioning allows the sensors to detect premature aging states, since these nodes are located at the end of the most delay critical paths affecting memory response time and correct functional operation (Fig. 6).

The purpose of the sense amplifier is to detect the difference between signal transitions in the bit lines during read operation, in order to be able to generate a *pulsed signal*, with the pulse width being proportional to the delay of the bit line signal transition. The pulsed output signal of sense amplifier is then used as one of the clocked inputs of the aging sensors.

The transition speed of the bit-lines from low-to-high and from high-to-low has a tendency of becoming slower when the memory ages, or generally when a PVT (Process-Voltage-Temperature-Aging) degradation occurs. Thus, introducing delay for the sense-amp to produce the output signal.

Apart from aging sensor, a CMOS based configurable phase shifter is implemented to generate sensor specific clock signals derived from a single unified interface clock (CLK_{ori}). Due to limited space and document conciseness, the design of the phase shifter is not disclosed in this paper and will be presented in upcoming publications. The phase

shifter is a digitally controlled ($Ph_Ctrl[0:3]$) delay logic used to generate Clk_sens signal which in turn, is used to define the position of the left edge of SM window. The phase shifter provides 48ps maximum phase delay possible to control by 3ps deltas and 16 possible states according to the value of 4 bit Ph_Ctrl signal.

The CLK_{ori} signal is used as interface clock for both the memory unit and as a reference clock signal for the phase shifter. Moreover, CLK_{ori} is used to drive FF1 of the aging sensor (Fig. 2), determining the position of the right edge of SM window. CLK_{ei} is the inverted derivative of CLK_{ori} and is used to trigger the error indication status (EI) by driving FF3.

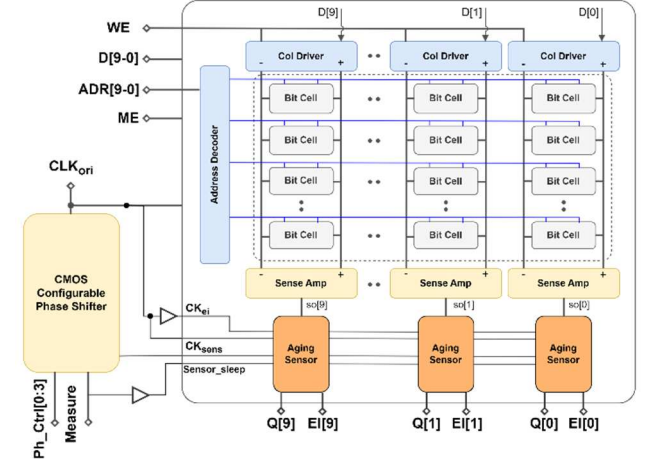


Fig. 6. FINFET-Based SRAM architecture with applied aging sensors

A. Sensor Bring-up and Calibration

Once the memory IP is manufactured, an initial calibration stage is required to determine the initial position of the CLK_{sens} signal. In more detail, to define the desired aging threshold for the fresh circuit. The purpose of the initial calibration is to determine the leftmost edge of the SM window for which none of the output nodes of the (fresh) memory violate SM window and no error indication is generated. The calibrated value of Ph_Ctrl control signal then is stored and used for each subsequent measurement cycle.

B. Sensor Measure and Idle Modes

An additional input pin is introduced (*Measure*) to control the measurement and idle cycles of the aging sensor. This signal controls phase shifter as well as the aging sensor through the *sensor_sleep* input. In measurement mode, when *Measure* signal is asserted, the phase shifter generates CLK_{sens} and CLK_{ei} clocks and the aging sensor is in functional mode. De-asserting *Measure* turns off CLK_{sens} , CLK_{ei} toggling and the aging sensor enters in sleep mode. This functionality not only helps to prevent unnecessary aging for phase shifter and sensor circuits, but also assures low power overhead when memory is not being monitored.

IV. RESULTS AND DISCUSSION

This section presents the obtained results of SPICE simulations demonstrating the capability of the proposed approach to detect memory performance degradation prior to the critical point when functional error is observed.

Fig. 7 shows standalone write/read operation of the fresh memory unit. Particularly, when *WE* signal is logical high, “1” bit is written on LSB (least significant bit) of the register with 4'b0 address. Then a read operation is performed, when *WE* is logical low. Bit line comparison leads to pulse generation on sense amplifier output node (*sens_amp_out[0]*), which is then captured by *CLK_ori* signal. Thus triggering *LSB* bit of the primary digital output (*q_out[0]*). The width of generated pulse signal by the sense amplifier is 64ps. In order for the generated pulse by sense amplifier to be correctly captured by output register, the required setup and hold times around the rising edge of *CLK_ori* must be satisfied, which basically defines the critical performance degradation level of the memory.

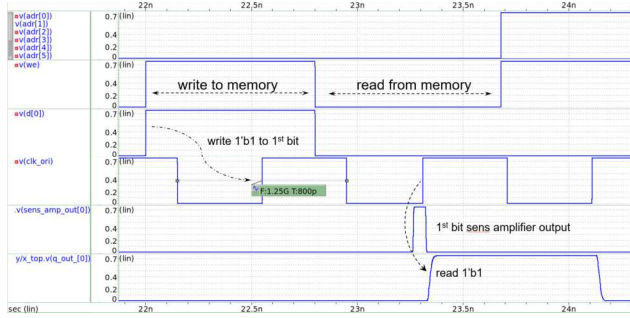


Fig. 7. Write and read operation simulation waveforms for fresh SRAM memory unit.

Fig. 8 shows the operation of aging sensor operation during “1” bit signal write and read operation into the fresh memory. As observed, *CLK_sense* is calibrated to form a 34ps SM window with respect to the rising edge of the *CLK_ori* signal. The defined SM window is not violated by the pulsed output signal of sense amplifier, leading the output of XOR cell of the sensor to be logical low-level during error indication capture cycle. In the sequence, the XOR output is captured by the rising edge of *CLK_ei* signal (FF3), which for this condition does not indicate error on *LSB* line.

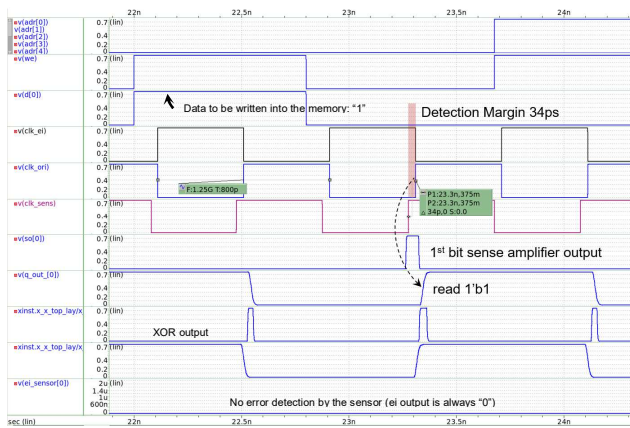


Fig. 8. Write and read operation simulation waveforms of fresh SRAM memory and applied aging sensor.

Fig. 9 shows the sensor operation with an aged memory unit, when aging leads to an introduction of additional delay on sense amplifier outputs, but does not lead to memory functional failure, i.e. still the correct data is read by the output register. In this case, *sense_amp_out[0]* signal violates the defined SM window of aging sensor, which is detected by *CLK_sens* rising edge. This condition leads

XOR to produce logical high-signal indicating a detected mismatch between captured signal values of FF1 and FF2.

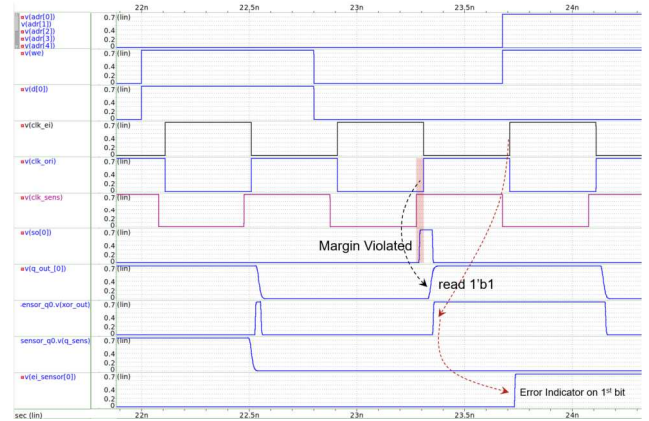


Fig. 9. Write and read operation simulation waveforms of aged SRAM memory and applied aging sensor.

When the rising edge of FF3 is asserted, this FF captures the XOR output and the *ei* output of the sensor is raised to “1”, indicating that the SM was violated for the *LSB* bit. The described case reveals the primary advantage of using the proposed approach, as user is notified about performance degradation due to aging prior to circuit functional failure, and thus, providing enough time to take counteract actions. This functionality is crucial for automotive and data center applications, where availability is always at a premium, and failure probability should be minimized to the lowest possible values.

An important metric of the sensor is the detection sensitivity, i.e., the minimum delay degradation that can be detected. During SPICE simulations, the minimum introduced delay on sense amplifier output node that is successfully detected by sensor is 4ps, which demonstrates high precision detection capabilities of proposed approach.

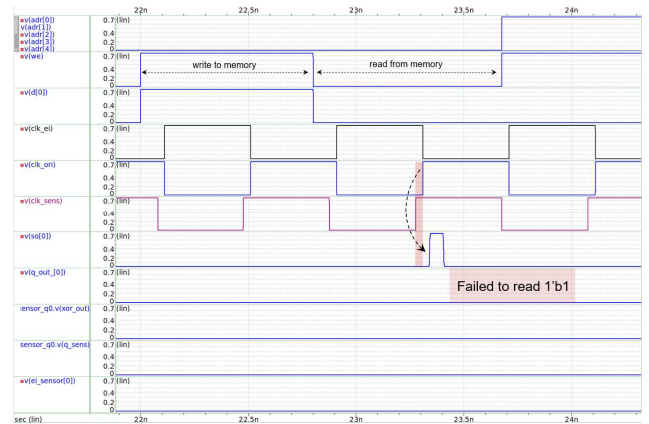


Fig. 10. Write and read operation simulation waveforms of aged SRAM memory and applied aging sensor.

Fig. 10 shows the critical situation when the memory aging state leads to a failure to properly capturing the data during a read operation. Here, the generated pulse by sense amplifier is shifted outside from the SM window, as well as setup and hold timing requirements are not satisfied for the rising edge of output register’s *CLK_ori* signal. As observed, the aging sensor will not detect the violation and error indication will not be asserted for this case. As the sensor is intended to detect degradation on the early stages

of aging when no functional failure occurs, this behavior is acceptable.

V. CONCLUSIONS

An effective on-chip aging sensor design is proposed to monitor performance degradation of FinFET-based SRAM IP core. The sensor presents as main characteristics the hardware simplicity: can be implemented with a few logic gates, and easy controllability: two additional clock signals should be generated and synchronized with respect to the original clock signal of the memory.

A case-study was implemented for a 5 nm SRAM IP core. It was verified by SPICE simulation that the sensor presents a very high sensitivity and is able to detect early aging states of the memory, before a functional error can be observed. Moreover, the sensor design features power gating capability by temporarily disconnecting it from power rails. So, in addition to prevent it from aging, this strategy also minimizes overall chip power consumption.

VI. DISCUSSIONS AND FUTURE WORK

This work presented an advanced, optimized design of an on-chip aging sensor. The sensor was implemented with a small number of logic gates (three DFFs and one XOR, if the sensor is placed in parallel with the original DFF at the output of the circuit critical path). This solution is interesting if the designer wishes to take advantage of the power gating technique (to prevent aging of the sensor, itself). Alternatively, if the sensor replaces the original circuit DFF, only two DFFs and the XOR are necessary (but in this case, the power gating approach cannot be used). In any case, the use of the sensor results in negligible overhead in terms of area, power and delay. Experimental results based on extensive SPICE simulations demonstrate that this approach is able to detect even the slight increase in memory response time due to aging much earlier than a functional error is observed at the memory output. This solution enables mission-mode monitoring of memory operation, which is a critical aspect of silicon lifecycle management (SLM) framework.

Our future work is built around the following three directions: (1) Conduction of extensive SPICE and Monte-Carlo simulations for different signoff PVT corners. (2) Further improvement concerns the aging sensor clock signal (clk_sens) to support both *rising* and *falling* edges of the circuit original clock (clk_ori). This will assure that the aging sensor can be directly coupled with memories controlled by either type clocked signal (*rising* and *falling*). (3) Another possible direction of this work could be the manufacturing of the memory IP block connected with the aging sensor as proposed herein. Then, apply high-temperature operating life (HTOL) test [19] to the chip and measure aging evolution of the memory and sensor, while determining how the sensor sensitivity is affected by aging.

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